

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Ecological monitoring & new technologies - Set 1

Mark scheme

AQA spec 3.1.2.3.5-3.1.2.3.6; The living environment; textbook pp. 402-407

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Size: shows whether the population is viable / how urgent action is; sets a baseline to judge success; needed to estimate carrying capacity and sustainable off-take • Distribution: shows which habitats / areas are used and whether the population is fragmented into non-viable isolated groups needing corridors; identifies key areas to protect • Survival rate (mortality): identifies the life stage and causes of death (predation, disease, poaching) so management targets the limiting factor • Age structure: many young indicates a growing population; few young / ageing population indicates breeding failure or high juvenile mortality even if adults are numerous; k-selected species need breeding adults protected	8
01.2	• Direct counts from vehicles / aircraft / drones along transects, or camera traps with individual recognition • Sub-sample counts in known areas scaled up to the whole reserve; mark-release-recapture (Lincoln index) for smaller species	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 400-402; AQA spec 3.1.2.3.5.	

Question	Marking guidance	Marks
02.1	- Tracking: continuous data on movements, territory size, migration routes and habitat use without disturbance, over large areas and long periods; acoustic tags in seawater; geolocators for small birds; more representative than occasional sightings - DNA: identifies individuals, gene pool size / inbreeding and regional populations (tracing ivory, timber, fish); eDNA detects rare or hidden aquatic species (great crested newt, invasive carp) from shed cells without capture - Image recognition: software identifies individuals from unique markings (tiger stripes, whale shark spots, fin damage) in camera-trap / drone / satellite images, giving population size, lifespan, social structure from large image sets quickly and objectively - Acoustic: bat detectors, hydrophones and sonograms record species that are hard to see (bats, cetaceans, insects), giving presence, abundance and activity continuously, day and night	8
02.2	- Data collected continuously / over large areas / at night rather than at a few times and places a researcher can visit - Less disturbance so animals behave naturally; larger sample sizes; objective rather than subjective identification	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-406; AQA spec 3.1.2.3.6.	
03.1	B - radio waves do not travel through seawater; acoustic or satellite tags transmit position	1
03.2	- Shows survival, dispersal and breeding of released individuals so success can be assessed and future releases improved - Radio / GPS tags or collars; leg rings / wing tags on birds; camera traps; DNA from droppings to confirm identity and diet	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 393-394, 402-407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
04.1	Environmental DNA (eDNA) analysis	1
04.2	- Transmitters are too heavy for small birds / their batteries would not last for a long-term study - Geocator tags are small and only record time and light levels; latitude and longitude for each day are worked out when the bird is recaptured and the data downloaded	2
04.3	- Each separate (regional) population has its own unique gene pool - DNA from seized ivory can be compared with the database to identify where the elephant came from, so poaching hotspots / trade routes can be targeted	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 404-405; AQA spec 3.1.2.3.6.	
05.1	- Images / maps of large areas; land-use change, deforestation, flooding - Vegetation density and surface temperature - Wind velocity, sea-surface altitude, wave height, oil pollution	3
05.2	Microwaves pass through clouds (unlike visible light)	1
05.3	- Very large / remote areas can be surveyed quickly and repeatedly so changes over time are detected - No need for researchers to visit inaccessible areas, reducing cost and disturbance; data are standardised across the whole area	2

Question	Marking guidance	Marks
05.4	<i>Any 4 from:</i> • Motion-sensitive cameras placed along trails record animals passing, using visible light by day and infrared at night • Each tiger has a unique pattern of facial / body stripes • Images are compared with the database so individuals are recognised rather than counted twice • The number of different individuals recorded gives a minimum population; repeated captures give territory size, movements, lifespans and social groupings • Limitation: only works for species with unique visible features	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-403, 406-407; AQA spec 3.1.2.3.6.	
06.1	• Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable • If it cannot be adjusted the percentage error should be recorded so the results can be corrected	2
06.2	<i>Any 3 from:</i> • Droppings: diet, gender, territory size • Owl pellets: diet • Feeding marks (eg chewed hazel nuts): presence of species such as dormice • Nests / burrows: presence and breeding • Tracks / footprints: presence, movements • Territorial marks (scratching posts): presence and territory	3
06.3	• Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species • The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen	2
06.4	<i>Any 3 from:</i> • Marked individuals mix freely with the rest of the population • Marks do not fall off or wear away, and marking does not affect survival • The population does not change during the study (no births, deaths or migration) • All individuals have an equal chance of being caught / animals are mobile and can be caught	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	